

Identifying errors in proofs - Set 1 - Solutions

Specification Err1, Err2.

Answer key

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Worked solutions

Question 1 ERR-01 · Identifying errors in proofs · spec Err2

A student attempts to solve the equation $\sqrt{x+3} = x-3$ for real x .

I Squaring both sides: $x+3 = (x-3)^2 = x^2 - 6x + 9$.

II So $x^2 - 7x + 6 = 0$.

III So $(x-1)(x-6) = 0$.

IV Hence the solutions are $x = 1$ and $x = 6$.

Which of the following best describes the solution?

Answer: A

Squaring can introduce extraneous solutions, so line IV should check them: $x = 1$ gives $\sqrt{4} = 2 \neq -2$. Only $x = 6$ works. Lines I-III are valid deductions.

Full method - how and why

Squaring an equation is not reversible; it can create solutions that do not satisfy the original. The error is the **missing check**, not the algebra.

1 Check line I: squaring both sides of $\sqrt{x+3} = x-3$ gives $x+3 = x^2 - 6x + 9$. Valid.

why: $A = B \Rightarrow A^2 = B^2$ is always a legitimate deduction - it is the converse that fails.

2 Check lines II and III: $x^2 - 7x + 6 = 0$ factorises as $(x-1)(x-6) = 0$. Valid.

why: $-1 \times -6 = 6$ and $-1 - 6 = -7$ ✓.

3 Line IV asserts both are solutions. Substitute $x = 1$ into the **original**: $\sqrt{4} = 2$, but $x-3 = -2$. $2 \neq -2$, so $x = 1$ is extraneous.

why: A square root is non-negative, so the right-hand side must be ≥ 0 , forcing $x \geq 3$.

4 Substitute $x = 6$: $\sqrt{9} = 3$ and $6-3 = 3$ ✓. So only $x = 6$ is a solution, and the first error is line IV.

why: Lines I-III produce **candidates**; the check is a required final step.

Where it goes wrong

- Blaming line I - squaring is a valid implication, just not an equivalence.
- Declaring the solution correct because the factorising is right; correct algebra can still give a wrong answer set.
- Checking the candidates in the squared equation rather than the original - both pass there, which proves nothing.

Takeaway After squaring, always substitute back into the **original** equation; squaring can only add solutions, never lose them.

Question 2 ERR-02 · Identifying errors in proofs · spec Err2

Here is a purported proof that $1 = 2$.

- I Let $a = b$, with $a \neq 0$.
- II Then $a^2 = ab$.
- III So $a^2 - b^2 = ab - b^2$, i.e. $(a - b)(a + b) = b(a - b)$.
- IV Dividing both sides by $(a - b)$: $a + b = b$.
- V So $2b = b$, and hence $2 = 1$.

On which line does the first error occur?

Answer: A

Since $a = b$, $a - b = 0$ and dividing by zero is invalid.

Full method - how and why

The classic ' $1 = 2$ ' fallacy. Somewhere a step is division by an expression that is secretly zero.

- 1 Line I sets $a = b$ with $a \neq 0$ - a legitimate supposition.
why: Nothing wrong yet.
- 2 Line II: multiplying $a = b$ by a gives $a^2 = ab$. Valid.
why: Multiplying both sides by the same quantity is always allowed.
- 3 Line III: subtracting b^2 from both sides and factorising both sides. Valid.
why: $a^2 - b^2 = (a - b)(a + b)$ and $ab - b^2 = b(a - b)$ - correct algebra.
- 4 Line IV divides by $(a - b)$. But $a = b$, so $a - b = 0$: this is division by zero. **First error.**
why: Dividing by zero is undefined, and it is exactly what manufactures the false conclusion.

Where it goes wrong

- Blaming line V, where the absurdity finally appears; the last line is where you **notice** the error, not where it happens.
- Blaming line III - the factorisation is perfectly correct.
- Being reassured by the stated condition $a \neq 0$; it is $a - b$, not a , that vanishes.

Takeaway In every fake proof of this type, look for a division by a factor that the hypotheses force to be zero.

Question 3 ERR-03 · Identifying errors in proofs · spec Err2

A student tries to solve $\sin \theta = \sin 40^\circ$ for $0^\circ \leq \theta < 360^\circ$.

I Since $\sin \theta = \sin 40^\circ$, we have $\theta = 40^\circ$.

II The sine function has period 360° , so $\theta = 40^\circ + 360^\circ k$.

III In the given interval the only solution is $\theta = 40^\circ$.

Which of the following is correct?

Answer: A

'If $\sin \alpha = \sin \beta$ then $\alpha = \beta$ ' is a classic invalid deduction; $\sin(180^\circ - 40^\circ) = \sin 40^\circ$ too.

Full method - how and why

$\sin$ is not one-to-one, so $\sin \theta = \sin \alpha$ has more solutions than $\theta = \alpha$.

1 Line I claims $\sin \theta = \sin 40^\circ \Rightarrow \theta = 40^\circ$. This is the error.

why: Sine takes the same value twice per revolution: $\sin(180^\circ - \alpha) = \sin \alpha$.

2 So $\theta = 140^\circ$ also satisfies the equation.

why: $\sin 140^\circ = \sin 40^\circ$ - check on the graph or with the symmetry of the unit circle.

3 Lines II and III are then correct **statements** in themselves (sine does have period 360°), but they are built on the incomplete line I.

why: The question asks for the **first** error, which is line I.

4 The full solution set in $0^\circ \leq \theta < 360^\circ$ is $\theta = 40^\circ$ and $\theta = 140^\circ$.

why: Only two, because the sine is positive in the first and second quadrants.

Where it goes wrong

- Claiming the period of sine is 180° - that is tangent.
- Claiming $\theta = 400^\circ$ was missed - it is outside the stated interval.
- Declaring the work correct because each individual line looks plausible.

Takeaway $\sin \theta = \sin \alpha$ gives $\theta = \alpha + 360^\circ k$ **or** $\theta = 180^\circ - \alpha + 360^\circ k$. **Never cancel a sine.**

Question 4 ERR-04 · Identifying errors in proofs · spec Err1

Claim: for all real x , if $x^2 > 9$ then $x > 3$.

- I Suppose $x^2 > 9$.
- II Taking square roots of both sides, $x > 3$.
- III This proves the claim.

Which of the following is true?

Answer: A

$x^2 > 9$ gives $|x| > 3$, i.e. $x > 3$ or $x < -3$. The claim itself is false, and line II is where the invalid deduction happens.

Full method - how and why

Taking square roots of an inequality gives a modulus, not a bare inequality. Here the claim itself is false.

- 1 Test the claim: $x = -4$ gives $x^2 = 16 > 9$, but $x = -4 > 3$. The **claim is false**.
why: Settle the truth of the claim first; that tells you an error must exist somewhere.
- 2 Line I is just the supposition - no error.
why: You are allowed to suppose the hypothesis.
- 3 Line II says 'taking square roots, $x > 3$ '. What $x^2 > 9$ actually gives is $|x| > 3$, i.e. $x > 3$ **or** $x < -3$. **First error**.
why: $\sqrt{x^2} = |x|$, not x .
- 4 The offered repair ('say $|x| > 3$ and then deduce $x > 3$ ') is no repair at all, since $|x| > 3$ does **not** give $x > 3$.
why: That option admits the modulus but then makes the same invalid step.

Where it goes wrong

- Accepting the option that says the claim is true - it is not.
- Blaming line III; by then the damage is done.
- Solving $x^2 > 9$ as $-3 < x < 3$, which is the solution of $x^2 < 9$.

Takeaway $x^2 > k^2 \Leftrightarrow |x| > k \Leftrightarrow x > k$ or $x < -k$. Squaring and rooting both lose sign information.

Question 5 ERR-09 · Identifying errors in proofs · spec Err2

A student proves that the equation $x^2 + 2x + 3 = 0$ has no real solutions.

- I Complete the square: $x^2 + 2x + 3 = (x + 1)^2 + 2$.
- II $(x + 1)^2 \geq 0$ for all real x .
- III So $(x + 1)^2 + 2 \geq 2 > 0$ for all real x .
- IV Hence the equation has no real solutions.

Which of the following is correct?

Answer: A

Every step is valid.

Full method - how and why

Another valid proof. Completing the square and bounding below is a standard, correct technique.

- 1 Line I: $(x + 1)^2 = x^2 + 2x + 1$, so $x^2 + 2x + 3 = (x + 1)^2 + 2$ ✓.
why: Half the coefficient of x is 1; correct by $3 - 1 = 2$.
- 2 Line II: any real square is ≥ 0 ✓.
why: True for every real x .
- 3 Line III: adding 2 gives ≥ 2 , and $2 > 0$ ✓.
why: Adding a constant to both sides of an inequality is always valid.
- 4 Line IV: an expression that is always at least 2 can never equal 0, so there are no real roots ✓.
why: That is the definition of having no real solutions. (Check: discriminant $4 - 12 = -8 < 0$ ✓.)

Where it goes wrong

- Claiming $(x + 1)^2$ can be negative - not for real x .
- Claiming 'a positive expression can still equal zero' - it cannot; that is what positive means.
- Assuming there must be an error.

Takeaway Completing the square gives a rigorous lower bound; if the minimum is positive, there are no real roots.

Question 6 ERR-10 · Identifying errors in proofs · spec Err2

Claim: if $a > b$ then $\frac{1}{a} < \frac{1}{b}$.

I Suppose $a > b$.

II Dividing both sides by ab : $\frac{1}{b} > \frac{1}{a}$.

III So $\frac{1}{a} < \frac{1}{b}$.

Which of the following is correct?

Answer: A

$a = 1, b = -1$: $a > b$ but $\frac{1}{a} = 1 > -1 = \frac{1}{b}$. Dividing by ab without knowing its sign is invalid (and $b = 0$ would be division by zero).

Full method - how and why

The same unknown-sign trap as ERR-05, now with division, plus a false claim.

1 Test the claim: $a = 1, b = -1$. Then $a > b$ but $\frac{1}{a} = 1$ and $\frac{1}{b} = -1$, so $\frac{1}{a} > \frac{1}{b}$. **Claim false.**

why: Opposite signs are the natural test for a reciprocal claim.

2 Line I is just the supposition - fine.

why: No deduction yet.

3 Line II divides by ab and keeps the direction. That needs $ab > 0$, which is not given; ab could be negative (as in the counterexample) or zero. **First error.**

why: Dividing by a negative reverses the inequality, and dividing by zero is undefined.

4 The claim is true if a and b have the same sign - which is the condition that was silently assumed.

why: Naming the missing hypothesis is the cleanest way to describe this kind of error.

Where it goes wrong

- Accepting the argument because it 'works for positive numbers' - the claim is stated for all reals.
- Blaming line III, which is a trivial restatement of line II.

Takeaway Reciprocals reverse order only within a fixed sign: for $a, b > 0$ (or both negative), $a > b \Rightarrow \frac{1}{a} < \frac{1}{b}$.

Question 7 ERR-12 · Identifying errors in proofs · spec Err2

A student solves $x^2 = 4x$.

I Divide both sides by x : $x = 4$.

II So the only solution is $x = 4$.

Which of the following is correct?

Answer: A

Factorise instead: $x(x - 4) = 0$ gives $x = 0$ or $x = 4$.

Full method - how and why

Dividing an equation by a variable destroys the root where that variable is zero.

1 Line I divides $x^2 = 4x$ by x . That is only legitimate when $x \neq 0$ - and $x = 0$ does satisfy the equation. **First error.**

why: $0^2 = 4 \times 0$ ✓, so $x = 0$ is a genuine solution that the division removes.

2 The correct method: bring everything to one side and factorise. $x^2 - 4x = 0$, so $x(x - 4) = 0$.

why: Factorising preserves every root, because a product is zero exactly when a factor is.

3 So $x = 0$ or $x = 4$ - two solutions.

why: Line II's 'only solution' is wrong as a consequence of line I.

Where it goes wrong

- Blaming line II - it faithfully reports what line I produced; the first error is the division.
- Thinking $x^2 \div x = 2x$; it is x .
- Trying to 'fix' it by remembering to add $x = 0$ afterwards - correct in this case, but factorising is the reliable habit.

Takeaway Never divide an equation by a variable expression. Factorise instead: division loses roots, squaring adds them.

Question 8 ERR-14 · Identifying errors in proofs · spec Err2

Claim: if x and y are real with $x^2 = y^2$, then $x = y$.

Which of the following is a counterexample to the claim?

Answer: C

A counterexample must satisfy the hypothesis ($x^2 = y^2$) and fail the conclusion ($x \neq y$).

Full method - how and why

A counterexample must satisfy the hypothesis and break the conclusion. Test both halves on each candidate.

1 Hypothesis: $x^2 = y^2$. Conclusion to break: need $x \neq y$.

why: Both conditions, or it is not a counterexample.

2 $x = 2, y = -2$: $4 = 4$ ✓ hypothesis, and $2 \neq -2$ ✓ conclusion fails. **Counterexample.**

why: Equal squares with opposite signs is exactly the gap in the claim.

3 $x = 2, y = 2$ and $x = 0, y = 0$ and $x = -1, y = -1$: hypothesis holds but so does the conclusion - these are examples of the claim.

why: Confirming instances never refute.

4 $x = 1, y = 2$: $1 \neq 4$, so the hypothesis fails and the case is irrelevant.

why: A case outside the hypothesis says nothing.

Where it goes wrong

- Picking a pair where the hypothesis fails and thinking the different values refute the claim.
- Picking a pair where both hold, which supports rather than refutes.

Takeaway $x^2 = y^2 \Rightarrow x = \pm y$. A counterexample satisfies the 'if' and fails the 'then'.

Question 9 ERR-16 · Identifying errors in proofs · spec Err2

A student argues: 'Every prime number is odd, because if it were even it would be divisible by 2 and so not prime.' Which of the following is correct?

Answer: A

A number divisible by 2 is not prime unless it equals 2. The conclusion 'every prime is odd' is false.

Full method - how and why

An argument that proves 'almost' the right thing. Find the single case it mishandles.

- 1 The reasoning is: $\text{even} \Rightarrow \text{divisible by 2} \Rightarrow \text{not prime}$.
why: Trace the chain and test each link.
- 2 The second link fails for one number: 2 is divisible by 2 and **is** prime, because divisibility by itself is allowed.
why: A prime has exactly two divisors, 1 and itself; for 2 those are 1 and 2.
- 3 So the conclusion 'every prime is odd' is false - 2 is an even prime.
why: A single counterexample refutes it.
- 4 The correct statement is 'every prime **greater than 2** is odd', and the argument does establish that.
why: Identifying the corrected claim shows precisely what the flaw was.

Where it goes wrong

- Objecting that some odd numbers are not prime - true, but irrelevant; the claim is not the converse.
- Objecting via 1 or 0 - neither is a counterexample to 'every prime is odd'.

Takeaway Check whether an argument's chain breaks at a boundary case. 2 is the even prime that breaks this one.

Question 10 ERR-18 · Identifying errors in proofs · spec Err2

A student solves $2\cos^2 x = \cos x$ for $0 \leq x < 2\pi$.

I Divide both sides by $\cos x$: $2\cos x = 1$.

II So $\cos x = \frac{1}{2}$.

III So $x = \frac{\pi}{3}$ or $x = \frac{5\pi}{3}$.

How many solutions has the student missed, and where is the first error?

Answer: A

Dividing by $\cos x$ loses $\cos x = 0$, i.e. $x = \frac{\pi}{2}$ and $x = \frac{3\pi}{2}$: two solutions.

Full method - how and why

Dividing a trigonometric equation by $\cos x$ loses the solutions where $\cos x = 0$.

- 1 Line I divides by $\cos x$, valid only if $\cos x \neq 0$. **First error.**
why: The equation $2\cos^2 x = \cos x$ is satisfied whenever $\cos x = 0$, so those solutions are discarded.
- 2 The correct method: $2\cos^2 x - \cos x = 0$, so $\cos x(2\cos x - 1) = 0$.
why: Factorise rather than divide - it keeps both branches.
- 3 Branch $\cos x = 0$: in $0 \leq x < 2\pi$ that is $x = \frac{\pi}{2}$ and $x = \frac{3\pi}{2}$ - **two** missed solutions.
why: Cosine vanishes at the top and bottom of the unit circle.
- 4 Branch $\cos x = \frac{1}{2}$: $x = \frac{\pi}{3}$ and $x = \frac{5\pi}{3}$ - the two the student found.
why: Cosine is positive in the first and fourth quadrants, so $x = \frac{\pi}{3}$ and $2\pi - \frac{\pi}{3}$. Four solutions in total.

Where it goes wrong

- Saying one solution was missed; $\cos x = 0$ has two solutions in a full revolution.
- Blaming line III - the student's own two values are correct for their (incomplete) equation.
- Declaring the working correct because the two values found do satisfy the original equation.

Takeaway Never divide a trigonometric equation by a trig function - factorise, and count the solutions of **each** branch across the whole interval.

Question 11 X2-25 · Identifying errors in proofs · spec Err2

A student solves $x^2 - 5x + 6 > 0$ as follows.

I $x^2 - 5x + 6 = (x - 2)(x - 3).$

II So $(x - 2)(x - 3) > 0.$

III So $x - 2 > 0$ and $x - 3 > 0.$

IV So $x > 3.$

Which of the following is correct?

Answer: A

The full solution set is $x < 2$ or $x > 3$.

Full method - how and why

A quadratic inequality solved as if it were an equation-style 'both factors positive'. Find where the missing branch is lost.

1 Lines I and II are fine: $x^2 - 5x + 6 = (x - 2)(x - 3)$, so the inequality is $(x - 2)(x - 3) > 0$.

why: Check: $-2 \times -3 = 6$ and $-2 - 3 = -5$ ✓.

2 Line III says both factors must be positive. But a product is positive when both are positive **or** both are negative. **First error.**

why: Two negatives multiply to a positive - the case the student omits.

3 The missing branch: $x - 2 < 0$ and $x - 3 < 0$, i.e. $x < 2$.

why: Both factors negative requires the smaller bound.

4 Full solution: $x < 2$ or $x > 3$. (Sketch check: the parabola is above the axis outside its roots.)

why: A quick sketch of an upward parabola confirms the two outer intervals.

Where it goes wrong

- Blaming line IV - it correctly follows from line III's (incomplete) conditions.
- Giving the answer as $2 < x < 3$, which is the solution of < 0 , not > 0 .
- Solving factor-by-factor at all; sketching the parabola is faster and safer.

Takeaway $(x - p)(x - q) > 0$ holds **outside** the roots, < 0 **between** them. Sketch rather than case-split.

Question 12 ERR-05 · Identifying errors in proofs · spec Err2

A student solves the inequality $\frac{x+1}{x-2} < 3$.

I Multiply both sides by $(x-2)$: $x+1 < 3x-6$.

II So $7 < 2x$.

III So $x > \frac{7}{2}$.

Which of the following is correct?

Answer: A

For $x < 2$ the multiplier is negative and the inequality reverses; indeed $x = 0$ gives $-\frac{1}{2} < 3$, a solution missed by the student. The full solution set is $x < 2$ or $x > \frac{7}{2}$.

Full method - how and why

Multiplying an inequality by an expression whose sign is unknown. The error is the unstated assumption.

1 Line I multiplies by $(x-2)$ and keeps the direction of the inequality.

why: That is only valid if $x-2 > 0$; if $x < 2$ the inequality must reverse.

2 So line I is the **first error**: it silently assumes $x > 2$.

why: The line is not wrong for all x , but it is wrong as an unconditional step - which is what makes it an error.

3 Consequence: the student misses a whole branch. Test $x = 0$: $\frac{0+1}{0-2} = -\frac{1}{2} < 3$ ✓ - a genuine solution not in $x > \frac{7}{2}$.

why: A single test value exposes the omission.

4 The correct solution set is $x < 2$ or $x > \frac{7}{2}$.

why: For $2 < x < \frac{7}{2}$ the fraction exceeds 3; at $x = 2$ it is undefined.

Where it goes wrong

- Saying you can never multiply an inequality by an expression - you can, if you case-split on its sign, or multiply by $(x-2)^2$, which is always positive.
- Blaming line II or III; the arithmetic there is right, given line I.
- Forgetting that $x = 2$ must be excluded from any solution set.

Takeaway Never multiply an inequality by an expression of unknown sign: split into cases, or multiply by its square.